WORKED SOLUTION

STEP I 1998, Question 8

Pure • Fluid dynamics

> **STEP I 1998, Question 8**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/98-s1-q8

The key idea

Despite the fluid-dynamics dressing, this is a pure differential equations question. The left-hand side is a derivative of a derivative, so we just integrate twice, remembering that dividing by r in between turns a constant into a logarithm. Physical sense, a finite velocity on the axis, then pins down one constant, and the no-slip condition at the wall pins down the other.

Solution

The general solution

Multiplying the equation by r gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}v}{\mathrm{d}r}\right) = -kr.$$

Integrating once with respect to r,

$$r\frac{\mathrm{d}v}{\mathrm{d}r} = -\tfrac{1}{2}kr^2 + A,$$

where A is a constant. We divide by r (fine, since $r > 0$ inside the tube away from the axis):

$$\frac{\mathrm{d}v}{\mathrm{d}r} = -\tfrac{1}{2}kr + \frac{A}{r}.$$

Integrating again, and using $\ln r$ rather than $\ln|r|$ because $r > 0$,

$$\boxed{v = -\tfrac{1}{4}kr^2 + A\ln r + B}$$

with A and B arbitrary constants. Two constants is what we expect for a second-order equation.

Behaviour near the axis

As $r \to 0^+$, the terms $-\frac{1}{4}kr^2$ and B stay bounded (they tend to 0 and B), while $\ln r \to -\infty$. So if $A \neq 0$, then $A \ln r$ tends to $-\infty$ or $+\infty$ (according to the sign of A), and hence $|v| \to \infty$. The only way to avoid this is to take $A = 0$, in which case $v \to B$, which is finite.

Ask yourself Why should we insist that $A = 0$?

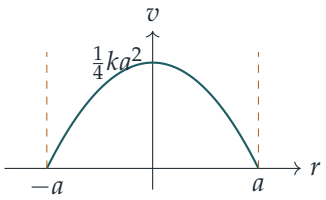
The axis $r = 0$ is inside the fluid, so the speed there must be finite: real fluid doesn't flow infinitely fast along the centre of a pipe. The $\ln r$ solution is a perfectly good solution of the equation for $r > 0$, but it's physically unacceptable here. (It would be allowed for flow between two coaxial cylinders, where the axis isn't part of the fluid.)

The velocity profile

With $A = 0$, we have $v = B - \frac{1}{4}kr^2$. The condition $v = 0$ at $r = a$ gives $B = \frac{1}{4}ka^2$, so

$$\boxed{v = \tfrac{1}{4}k\left(a^2 - r^2\right).}$$

The profile is a parabola: the speed is greatest, $\frac{1}{4}ka^2$, on the axis and falls to zero at the wall.



(The diagram shows a slice across a diameter, with the walls dashed.)

The flow rate

Substituting into the given formula,

$$F = 2\pi \int_0^a r \cdot \tfrac{1}{4}k\left(a^2 - r^2\right) \mathrm{d}r = \frac{\pi k}{2} \int_0^a \left(a^2 r - r^3\right) \mathrm{d}r$$
$$= \frac{\pi k}{2}\left[\frac{a^2 r^2}{2} - \frac{r^4}{4}\right]_0^a = \frac{\pi k}{2} \cdot \frac{a^4}{4},$$

so

$$\boxed{F = \frac{\pi k a^4}{8}.}$$

Checks

- Substitute $v = \frac{1}{4}k(a^2 - r^2)$ back: $r\,v' = -\frac{1}{2}kr^2$, whose derivative is $-kr$, and dividing by r gives $-k$, as required.
- The average speed is $F/(\pi a^2) = \frac{1}{8}ka^2$, exactly half the maximum speed $\frac{1}{4}ka^2$, which is the well-known property of a parabolic profile.
- Numerically, with $k = 2$ and $a = 1$, Simpson's rule gives $F \approx 0.785398$, matching $\pi/4$.

Common slips

- Forgetting the constant at the first integration, which loses the $A \ln r$ term and makes the "unless" part meaningless.
- Integrating $-kr$ to get $-\frac{1}{2}kr^2$ and then forgetting to divide by r before integrating again.
- Dropping the factor r inside the integral for F: the flow through a thin ring of radius r is proportional to its circumference $2\pi r$, not to a constant.

Beyond STEP

The operator $\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}}{\mathrm{d}r}\right)$ is the Laplacian ∇^2 in cylindrical coordinates for a function of r alone, so this question is really Poisson's equation $\nabla^2 v = -k$; the $\ln r$ solution is the two-dimensional analogue of the $1/r$ potential of a point charge. The result $F \propto a^4$ is Poiseuille's law: halving a pipe's radius cuts the flow by a factor of sixteen, which is why narrowed arteries matter so much. For the differential equations side, see Boyce and DiPrima, *Elementary Differential Equations*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.